

Urban GIS Portfolio Project

GIS-Based Analysis of Residential Quality and Population Density in Tehran, Iran

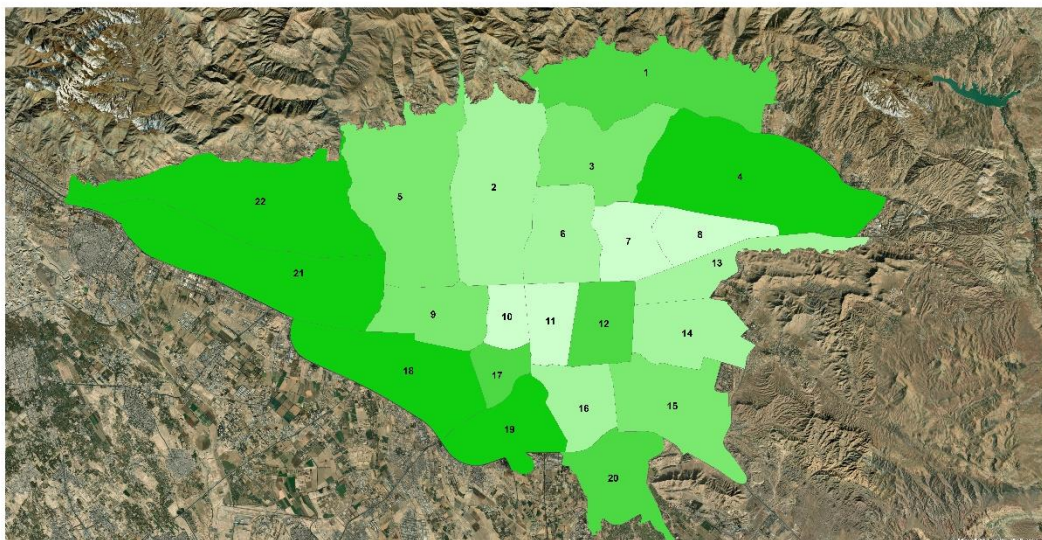
Master's Thesis (Tarbiat Modares University, Winter 2023–2024)

Case study: 22 Districts of Tehran, Iran

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Tools: ArcGIS | SPSS | Excel | Spatial Analysis

Residential Quality Index (RQI) of Tehran Districts



Legend

Residential Quality Index (RQI)

- Very Low
- Low
- Moderate
- High
- Very High

0 1.75 3.5 7 Miles

Prepared by: Samira Heydari
Source: Tehran Statistical Data
Software: ArcGIS

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Executive Summary:

This project examines how population density relates to the objective quality of the residential environment across Tehran's 22 districts. Using multi-dimensional urban data (social, economic, physical, housing, environmental) and GIS-based analysis, we created a composite Residential Quality Index. Key methods include ArcGIS mapping, Exploratory Factor Analysis (EFA), and statistical correlation. The study identifies spatial patterns of housing quality and informs data-driven urban planning.

Project Overview:

The study integrates official census and urban datasets to assess neighborhood conditions in Tehran. We compiled about 60 variables (demographics, economy, land use, housing, and environment) and mapped them at the district level. In GIS, these indicators were merged with a city district shapefile to create thematic maps (e.g. population density, land use). Exploratory factor analysis reduced the indicators to 11 key factors across five dimensions. The resulting factor scores were combined into an overall Residential Quality Index. Finally, we analyzed correlations between population density and each quality dimension. The portfolio below outlines the data sources, workflow, map outputs, and key results of this analysis.

Objectives:

- Evaluate spatial differences in objective residential quality across Tehran's districts.
- Develop a composite Residential Quality Index from multiple urban indicators.
- Analyze the relationship between population density and residential environment quality.
- Identify districts with relatively high or low housing quality for targeted urban planning.

Data & Indicators:

The research framework included approximately 60 urban indicators grouped into five main dimensions: social, economic, physical, housing, and environmental factors. These indicators were standardized and prepared for GIS-based spatial analysis and statistical modeling.

Physical & Urban Indicators:

- Land Use Patterns: residential, commercial, administrative, educational, healthcare, recreational, and green spaces distribution.

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- Per-Capita Urban Land Use: availability of urban facilities and services relative to population.
- Accessibility to Urban Services: access to public transportation, educational centers, healthcare facilities, cultural centers, and recreational spaces.

Housing & Residential Quality Indicators:

- Housing Ownership and Rental Rates
- Housing Unit Type and Average Unit Area
- Building Age Distribution
- Structural Building Type
- Deteriorated Urban Fabric (Urban Decay)
- Housing Condition Indicators

Environmental Indicators:

- Air Pollution Levels
- Green Space per Capita
- Environmental Quality Indicators

These indicators were integrated into GIS to evaluate spatial differences in residential quality across Tehran's 22 districts. In total, ~60 variables were used, categorized into these five dimensions. These variables were standardized and input into factor analysis.

Indicator (Persian)	Indicator (English)
جمعیت	Population
تراکم جمعیت	Population Density
بعد خانوار	Household Size
ساختار سنی	Age Structure
مهاجرت	Migration Rate
نرخ اشتغال	Employment Rate
نرخ مشارکت اقتصادی	Economic Participation Rate
نرخ بیکاری	Unemployment Rate
کاربری اراضی	Land Use
سرانه کاربری ها	Per-Capita Land Use
دسترسی به خدمات شهری	Access to Urban Services
حمل و نقل عمومی	Public Transportation Access

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Indicator (Persian)	Indicator (English)
مراکز آموزشی	Educational Facilities
مراکز درمانی	Healthcare Facilities
مراکز فرهنگی	Cultural Centers
فضاهای تفریحی	Recreational Spaces
مالکیت	Home Ownership
اجاره‌نشینی	Rental Housing
نوع واحد	Housing Unit Type
مساحت واحد	Housing Unit Area
قدمت بنا	Building Age
نوع اسکلت	Building Structure Type
بافت فرسوده	Deteriorated Fabric (Urban Decay)
آلاینده‌های هوا	Air Pollution Levels
شاخص‌های محیط‌زیستی	Environmental Quality Indicators

1. GIS Workflow:

Data Collection & Integration:

Official statistical data, census information, urban plans, land-use datasets, housing statistics, and environmental data were collected for Tehran's 22 districts.

2. GIS Database Development:

The Tehran district shapefile was imported into ArcGIS, and all statistical indicators were joined to the spatial database. Data layers were standardized using a common coordinate system.

3. Thematic Mapping & Spatial Visualization:

GIS-based thematic maps were generated to identify spatial patterns, including:

- Population Density Map
- Building Age Map
- Deteriorated Urban Fabric Map
- Residential Quality Index Map

4. Residential Quality Index Construction:

Approximately 60 indicators were analyzed using Exploratory Factor Analysis (EFA) in SPSS. Eleven key factors representing social, economic, physical, housing, and environmental dimensions were extracted. Factor scores were combined to create a composite Residential Quality Index.

5. Spatial Analysis & Evaluation:

The final quality index was classified into different categories (low, medium, high)

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and mapped. Correlation analysis was conducted to examine the relationship between population density and residential quality dimensions.

Methods:

- **Exploratory Factor Analysis (EFA):** Reduced 60 variables into 11 underlying factors representing dimensions of residential quality. Factors interpreted as social, economic, physical, housing, and environmental components.
- **Correlation Analysis:** Used Pearson's correlation in SPSS to test relationships between population density and the derived quality indices (overall and by dimension).
- **GIS Mapping:** Employed ArcGIS for spatial joins, map generation, and layout design. Data management and tables were handled in Excel.

Selected Maps:

1. Residential Quality Index Map

Spatial distribution of the composite Residential Quality Index across Tehran's 22 districts based on integrated socio-economic, physical, housing, and environmental indicators.

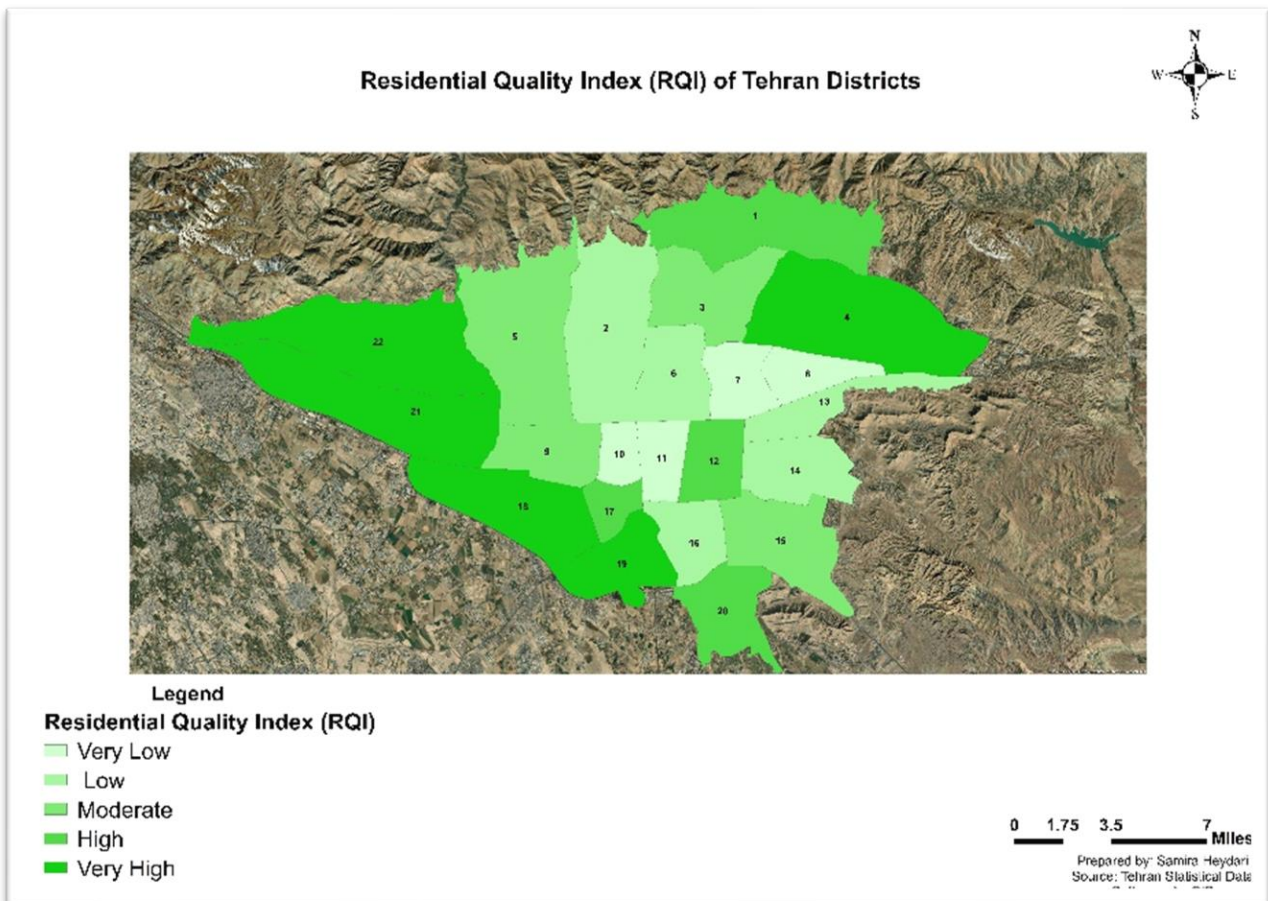


Figure 1. Residential Quality Index Map of Tehran's 22 Districts

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2. Population Density Map (2016–2021)

Spatial distribution of population density (persons per km²) across Tehran districts, showing variations in population concentration.

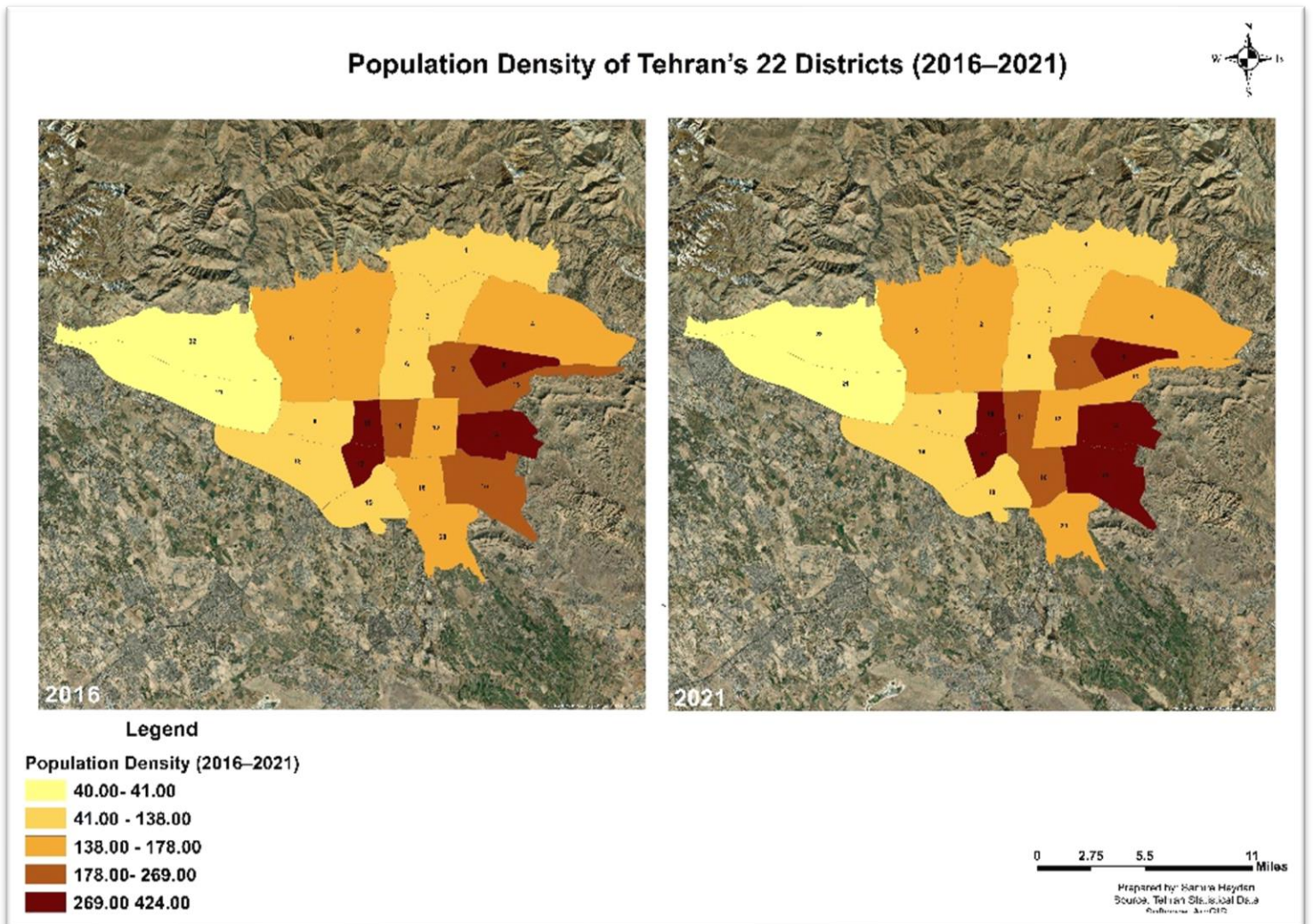


Figure 2. Population Density Map of Tehran's 22 Districts

3. Building Age Distribution Map

Spatial analysis of building age to identify districts with older residential structures and potential housing vulnerability.

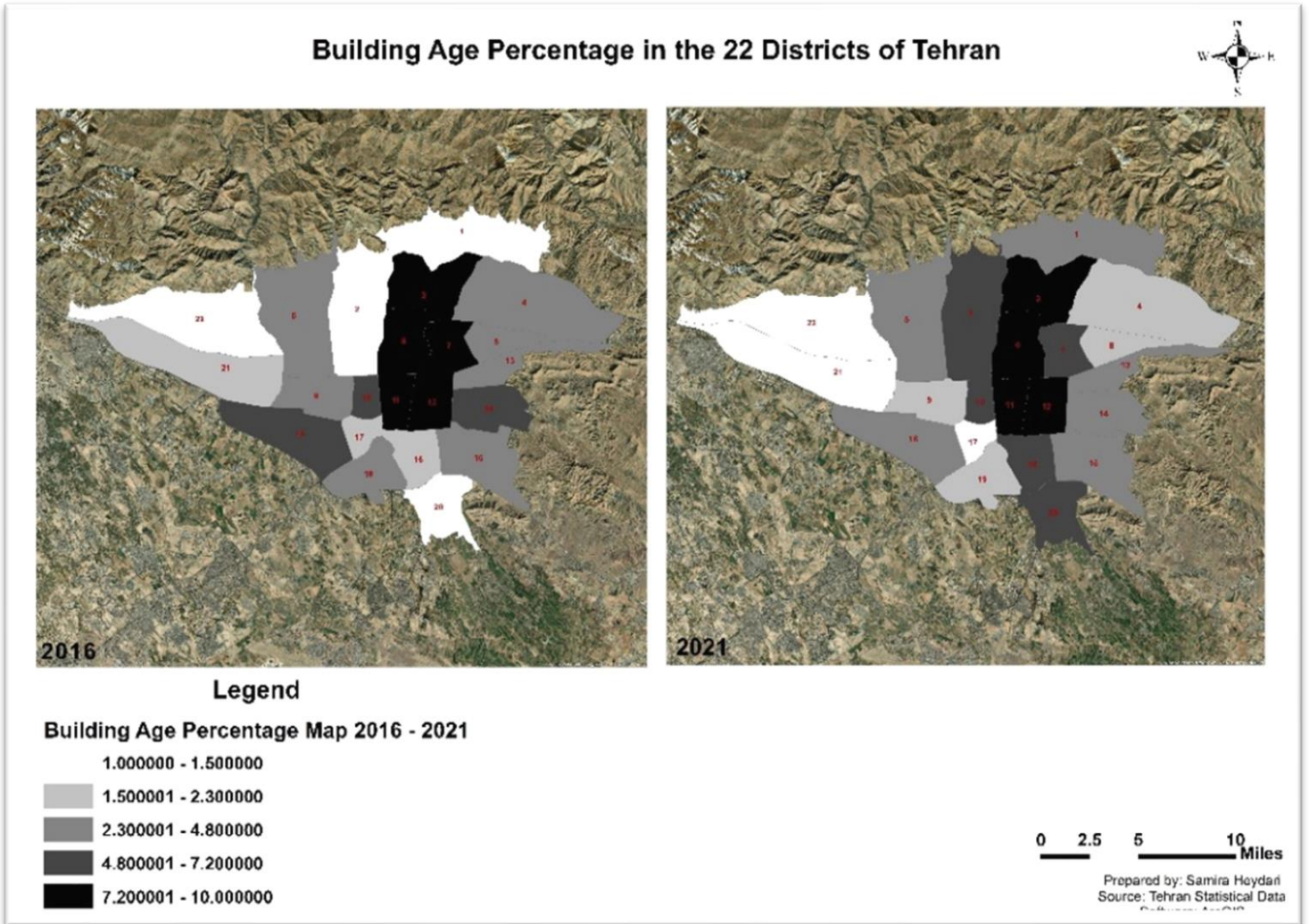
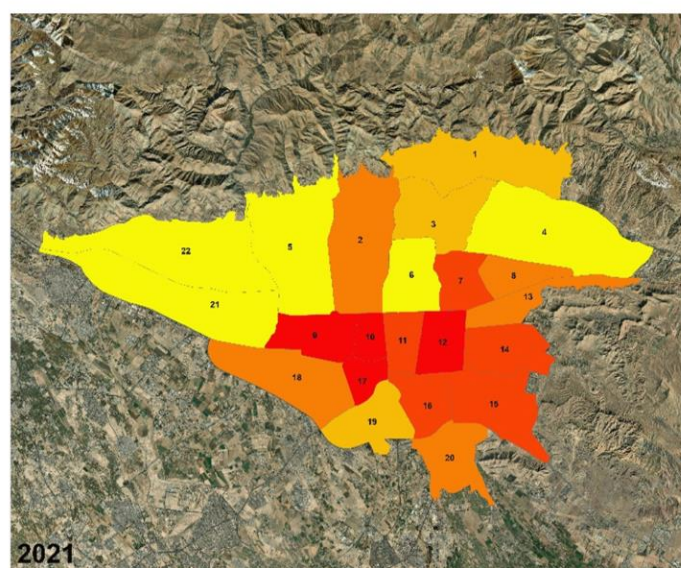
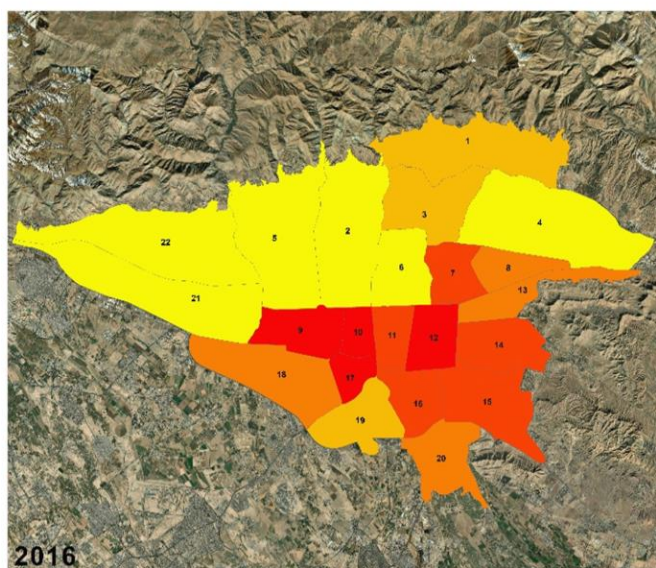


Figure 3. Building Age Percentage Map of Tehran's 22 Districts

4.Deteriorated Urban Fabric Map

Mapping of deteriorated and vulnerable urban areas to identify districts requiring regeneration and improvement strategies

Percentage of Deteriorated Urban Fabric in Tehran Districts (2016–2021)



Legend

Deteriorated Urban Fabric (%) 2016 - 2021



0 2.5 5 10 Miles

Prepared by: Samira Heydari
Source: Tehran Statistical Data
Software: ArcGIS

Figure 4. Deteriorated Urban Fabric Percentage Map of Tehran's 22 Districts

Key Findings:

- Districts **4, 6, 10, 19, 21, 22** had the lowest Residential Quality Index (below city average), indicating more disadvantaged housing conditions.
- Districts **2, 3, 7, 12, 14** scored highest on the quality index.
- The 60 input variables were reduced to 11 key factors (via EFA), capturing the main dimensions of urban quality.
- A significant positive correlation was found **only** between population density and the economic quality dimension. No significant correlation was observed between population density and other dimensions (social, housing, physical, environmental).
- The mapping highlights clear spatial patterns in housing quality, providing actionable insight for urban planners (e.g., targeting low-quality districts for improvement).

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• Skills Demonstrated:

- Spatial Data Analysis (ArcGIS Pro) – data integration, mapping, GIS database design
- Urban & Housing Geography – multi-dimensional urban indicator analysis
- Urban Spatial Analysis – land use evaluation, building age analysis, and housing vulnerability mapping
- Thematic Cartography & Data Visualization – clear map design with English annotations
- Statistical Analysis (SPSS) – factor analysis and correlation tests
- Excel Data Management – organizing and preprocessing large datasets
- Technical Reporting – concise documentation of GIS workflows and findings

CV/LinkedIn Summary:

GIS Analyst/Urban Planner with expertise in spatial data integration and thematic mapping. Master's thesis on Tehran's residential quality uses ArcGIS and statistical analysis to inform city-wide planning. Proven skills: geospatial analysis, factor modeling, data-driven reporting.

Attachments Checklist:

- ArcGIS project file (.aprx or .mxd) with data connections
- Tehran districts shapefile (and supporting GIS layers)
- Raw data tables (Excel/CSV of urban indicators)
- High-resolution map exports (**PNG/PDF**) of all selected maps
- Statistical output (SPSS tables or summary Excel) detailing factor analysis

Portfolio Finalization Timeline

